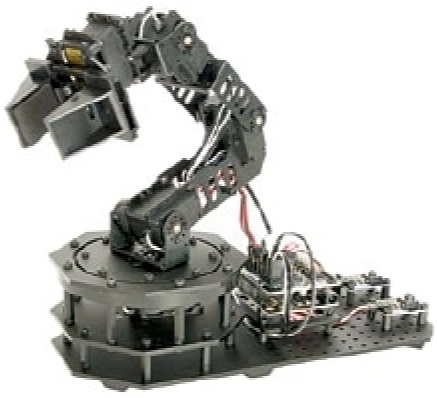

Fig. 3: Arduino-based robotic arm (Credit: www.trossenrobotics.com)

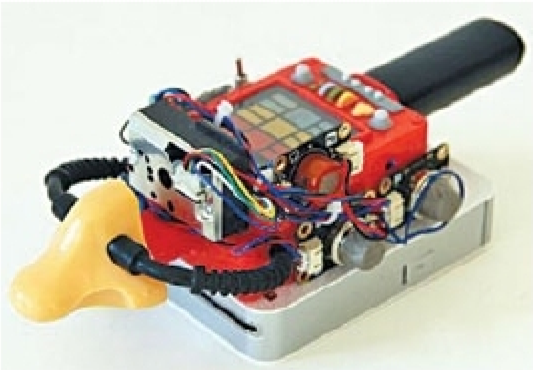

Fig. 4: Electronic nose module (Credit: https://blog.arduino.cc)

Fig. 5: Raindrop sensor (Credit: http://wiki.bernardino.org)

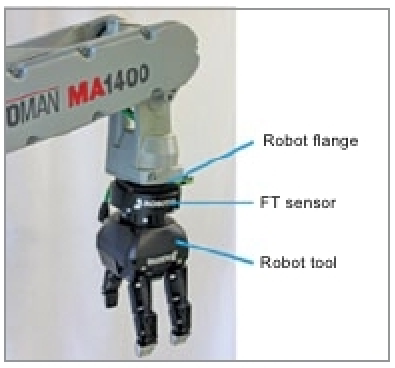

Fig. 6: 6-axis force torque sensor fitted onto robotic arm (Credit: https://blog.robotiq.com)

speech recognition and the like are achieved through sensory inputs such as cameras, microphones, wireless signals, active lidar, sonar and radar to deduce the different aspects of the world (including facial, object and gesture recognition).

Modern technology for AI robots can be a blessing or a disaster for human beings, at the same time. For example, a killer robot based on AI could be a very dangerous thing. There has been a huge campaign by AI experts all over the world to boycott killer robots reported to be under-development at Korea Advanced Institute of Science and Technology (KAIST). However, KAIST's president, Sung-Chul Shin, has denied any such development.

Robots are not only used for educational, industrial, military and civil applications, but also in movies and entertainment industry. Gravity, a science fiction thriller, used robots that were originally designed for performing assembly, welding and painting applications. The robotic arm driven by sensors was made to carry cameras, lights, props and even actors. By using the robotic arm behind the scenes, the director was able to achieve remarkable shots for the movie.

Then, there are robots such as QRIO and RoboSapien that are capable of advanced features like voice recognition or walking.

Sensors and robotics

A typical robot has a movable physical structure driven by motors, sensor circuitry, power supply and computer that controls these elements.

Some popularly known robots are:

1. Educational robots are mostly robotic arms used to lift small objects, or wheels to move forwards, backwards, left or right, with some other features.

2. Honda's ASIMO robot can walk on two legs like a person.

3. Industrial robots are automated machines that work on assembly lines.

4. BattleBots are remote-controlled fighters.

5. DRDO's Daksh is a battery-operated, remote-controlled robot on wheels. Its primary role is to recover bombs.

6. Robomow is a lawn-mowing robot.

7. MindStorms are programmable robots based on Lego building blocks.

Some robots only have motorised wheels, while others have dozens of movable parts, including spin wheels and pivot-jointed segments with some sort of actuators. Some robots use electric motors and solenoids as actuators, while others use a hydraulic or a pneumatic system. Most autonomous robots have sensors that can move and perform a specific task without human intervention or control.

One interesting technology under development in the field of robotics is the micro-robot for biological environment. A micro-robot may be integrated with sensors, signal processing, memory unit and feedback system working at a micro-scale level. This development can have important implications for integrated micro-bio-robotic systems for applications in biological engineering and research.

Some common robotic sensors are given in the table.

Biosensors. Biosensors use a living organism or biological molecules, especially enzymes or antibodies, to detect the presence of chemicals. Examples include blood glucose monitors, electronic noses and DNA biosensors.

Raindrops sensors. These sensors are used to detect rain and weather conditions, and convert the same into number of reference signals and analogue output.

Multi-axis force torque sensors. Such sensors are fitted onto the wrist of robots to detect forces and torques that are applied to the tool.

Optoelectronic torque sensors. These sensors are used for collaborative robot applications, ensuring safer and more effective human-machine-collaboration. These have less than one microvolt of noise even in low-torque range, can self-calibrate and measure to an accuracy of 0.01 per cent. A contactless measurement principle ensures that the sensors are insensitive to vibra-

COMMON SENSORS USED IN ROBOTS

Sensors	Functions
Touch	Sensing an object's presence or absence
Force	Measuring force along a single axis
Vision	Detecting edges, holes and corners
Proximity	Non-contact detection of an object
Physical orientation	Co-ordinates of objects in space
Heat	Wavelength of infrared (IR) or ultra violet (UV) rays, temperature, magnitude and direction
Chemicals	Presence, identity and concentration of chemicals or reactants
Light	Presence, colour and intensity of light
Sound	Presence, frequency and intensity of sound

Fig. 7: Optoelectronic torque sensor (Credit: www.dpaonthenet.net)